

Introductory Trigonometry Practice

Instructions

Complete in 40 minutes. Show clear work. Give exact values unless a rounding direction is stated. Use degrees or radians as shown.

Problems

1. Give the unit-circle coordinate $(\cos \theta, \sin \theta)$ for each angle:

$$\frac{5\pi}{6}, \quad \frac{7\pi}{4}, \quad 240^\circ, \quad -\frac{\pi}{3}.$$

2. Evaluate exactly:

$$\sin \frac{11\pi}{6}, \quad \cos \frac{3\pi}{4}, \quad \tan \frac{2\pi}{3}, \quad \sec \frac{5\pi}{3}, \quad \csc \frac{7\pi}{6}, \quad \cot \frac{3\pi}{4}.$$

3. A point on the unit circle is $(-\frac{12}{13}, \frac{5}{13})$. State the quadrant and find all six trigonometric function values.
4. In a right triangle, relative to angle θ , the opposite leg is 9 and the adjacent leg is 40. Find all six trigonometric function values.
5. A right triangle has an acute angle of 35° and hypotenuse 18. Find the opposite and adjacent side lengths to the nearest tenth.
6. From a point 28 feet from the base of a vertical building, the angle of elevation to the roof is 52° . Find the building's height to the nearest tenth of a foot.
7. Suppose $\sin \theta = \frac{3}{5}$ and $\cos \theta = -\frac{4}{5}$. Evaluate:

$$\begin{aligned} \sin(\theta + 4\pi), \quad \cos(\theta - 6\pi), \quad \tan(\theta + 5\pi), \\ \sec(\theta - 2\pi), \quad \csc(\theta + 8\pi), \quad \cot(\theta - 3\pi). \end{aligned}$$

8. State the period of each function:

$$\sin(3x), \quad \cos\left(\frac{x}{4}\right), \quad \tan(5x), \quad \cot\left(\frac{2x}{3}\right), \quad \sec(2x), \quad \csc\left(\frac{x}{3}\right).$$

9. A right triangle has hypotenuse 25 and one leg 7. Find the missing leg. If θ is opposite the missing leg, find $\sin \theta$, $\cos \theta$, and $\tan \theta$.

10. If $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III, find

$$\sin \theta, \quad \tan \theta, \quad \sec \theta, \quad \csc \theta, \quad \cot \theta.$$

11. Find each missing value using a Pythagorean identity and the stated quadrant.

(a) $\tan \alpha = \frac{7}{24}$ and α is in Quadrant I; find $\sec \alpha$.

(b) $\csc \beta = -\frac{5}{3}$ and β is in Quadrant IV; find $\cot \beta$.

12. Complete or evaluate each statement:

(a) $\sin 28^\circ = \cos(\text{_____}^\circ)$

(b) $\tan 17^\circ = \cot(\text{_____}^\circ)$

(c) $\sec 41^\circ = \csc(\text{_____}^\circ)$

(d) $\cos\left(\frac{\pi}{2} - \frac{\pi}{6}\right)$

(e) If θ is acute and $\sin \theta = \cos 23^\circ$, find θ .

Answer Key

1. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right), \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right), \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

2. $-\frac{1}{2}, -\frac{\sqrt{2}}{2}, -\sqrt{3}, 2, -2, -1$

3. Quadrant II;

$$\sin \theta = \frac{5}{13}, \quad \cos \theta = -\frac{12}{13}, \quad \tan \theta = -\frac{5}{12}, \quad \csc \theta = \frac{13}{5}, \quad \sec \theta = -\frac{13}{12}, \quad \cot \theta = -\frac{12}{5}.$$

4. The hypotenuse is 41;

$$\sin \theta = \frac{9}{41}, \quad \cos \theta = \frac{40}{41}, \quad \tan \theta = \frac{9}{40}, \quad \csc \theta = \frac{41}{9}, \quad \sec \theta = \frac{41}{40}, \quad \cot \theta = \frac{40}{9}.$$

5. Opposite ≈ 10.3 ; adjacent ≈ 14.7 .

6. ≈ 35.8 feet.

7. $\frac{3}{5}, -\frac{4}{5}, -\frac{3}{4}, -\frac{5}{4}, \frac{5}{3}, -\frac{4}{3}$

8. $\frac{2\pi}{3}, 8\pi, \frac{\pi}{5}, \frac{3\pi}{2}, \pi, 6\pi$

9. Missing leg 24; $\sin \theta = \frac{24}{25}, \cos \theta = \frac{7}{25}, \tan \theta = \frac{24}{7}.$

10. $\sin \theta = -\frac{12}{13}, \tan \theta = \frac{12}{5}, \sec \theta = -\frac{13}{5}, \csc \theta = -\frac{13}{12}, \cot \theta = \frac{5}{12}.$

11. (a) $\sec \alpha = \frac{25}{24}$ (b) $\cot \beta = -\frac{4}{3}$

12. (a) 62 (b) 73 (c) 49 (d) $\frac{1}{2}$ (e) 67°